

MATH 140 Fall 2026

notes by

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A note on these notes. This document will periodically be updated with the material that we cover. The timestamps in the left margin indicate when the notes from each day start. Subsections labelled with a * are optional. These notes should *not* be used as a substitute for the textbook or for attending class. (For instance, on the blackboard I will often draw graphs or diagrams, and these for the most part will be omitted from the notes.) The best way to do well in this class is to do as many exercises from the textbook as you can stomach. This document may contain errors, typographical or mathematical. Please email me if you find any.

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I. FOUNDATIONS

*Que si cependant la Geometrie a toujours
quelque obscurité essentiel, qu'on ne puisse dissiper,
& ce fera uniquement, à ce que je crois, du côté de l'Infini,
c'est que de ce côté-là la Geometrie tient à la Phisique,
à la nature intime des Corps que nous connoissons peu,
& peut-être aussi à une Metaphisique trop élevée,
dont il ne nous est permis que d'appercevoir quelques rayons.*

— BERNARD LE BOVIER DE FONTENELLE, *Éléments de la Geometrie de l'Infini* (1727)

1. Rates of change

31.VIII This set of notes assumes that the reader is familiar with all the pre-calculus concepts that one learns in the later years of secondary school: functions, trigonometry, exponential functions, logarithms, etc. However, given that the very definition of what differential calculus is requires us to talk about functions and their inputs, it is perhaps prudent to begin with a refresher on real-valued functions with real-valued outputs.

A *function* is a collection of ordered pairs (x, y) with the rule that if (x, y) and (x, z) are both in the collection, then $y = z$. We can view x as the *input* to the function, and y as the *output* of the function; under this view, the rule above is simply that the function never outputs two different values when given the same input. People like to use the letter f (or g or h , but really we can use any letter) to denote a function, and $f(x)$ to denote its value evaluated at a given input x . However, many particular functions are difficult to write without involving the input variable x in the expression, and in these cases we'll write $f(x)$ to mean the function as well.

Here are some examples of functions:

- a) The function $f(x) = 3x$, defined on all real numbers, that outputs triple its input value.
- b) The function that assigns to each real number x its square x^2 .
- c) The function $y = f(x)$ that equals 0 whenever x is a negative real number, and equals x whenever $x \geq 0$.
- d) The function that takes as input any nonzero real number x and outputs its reciprocal $1/x$.
- e) The function $f(x)$, defined on all real numbers, given by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational;} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Note that in each case above, we were careful to specify what the possible inputs to the function were. Going back to the definition of a function as a collection of ordered pairs (x, y) , the set of all x such that (x, y) is in the function is called the *domain* of f ; this is precisely the set of all valid inputs to the function f . (The set of all y such that (x, y) is in f is called the *range* of f , but this definition won't be as important to us.)

The title of this section is “rates of change,” but determining the rate at which a given function changes is actually what this *entire class* will essentially be about. We want to study how much the output of a function changes when you change its input.

For example, consider a car moving from Montreal to Quebec City. As time goes on, it gets further away from Montreal and closer to Quebec City. So we can model the trip as a function $d(t)$, where for each positive real number t we let $d(t)$ equal the distance of the car from Montreal (in kilometres), exactly t

minutes since it set off. The rate of change of this function at any given point in time t is the car's *velocity* at that point.

How do we calculate velocity? Well, we know that it is measured in units such as “kilometres per hour” or “metres per second,” so it is not too hard to see that velocity is calculated by dividing distance over time.

Let's go back to the car example. Suppose your car is equipped with a very good GPS that can measure your distance from Montreal; however, it does not have a working speedometer. Your friend (who is driving) asks you, “How fast am I driving?” You have taken high school physics, so you know how to solve this problem. You get out a stopwatch and start it exactly when the GPS measures your distance at 135.2 kilometres from Montreal. After one minute on the stopwatch, the GPS reads 137.4, so you report to your friend that she is driving $137.4 - 135.2 = 2.2$ kilometres per minute, or 132 kilometres per hour.

But your friend is not satisfied. “You told me my average speed over the course of a minute, not my speed at the moment I asked you the question. During that minute, I had to speed up to overtake a U-Haul and then slow down because someone cut me off in the fast lane.”

So you try and do better. You get out the stopwatch again, and when you start it the GPS reads 137.7. You wait ten seconds, then observe that the GPS reads 138.1. So you report your friend's new speed at $138.1 - 137.7 = 0.4$ kilometres every ten seconds, or 2.4 kilometres per minute, or 144 kilometres per hour.

Your friend is a little more satisfied, but wonders if you can do better. So you try one last time. You start the stopwatch when the GPS reads 138.5, wait exactly one second, and then check the GPS again. To everyone's surprise and disappointment, it still reads 138.5, because it only measures distance to the tenth part of a kilometre. So the calculation would report that the car has moved 0 kilometres in that one second, that is, it's going 0 kilometres per hour.

Obviously in this example you were hamstrung by the precision of your measuring instruments, but it is still worth pondering the following point. Your friend was asking for her speed at some particular instant. However, during any particular instant she does not move, nor does time move. So any naïve calculation as to her speed would go along the lines of $0/0$, which is an undefined value.

Yet it is undeniable that something's “speed at a particular instant” is a concept that holds meaning in everyday life. If you are in a plane and your friend is in a car below you, then it's certainly not nonsensical to say that you are moving faster than your friend at that moment, even though technically it could also be said that both of you have moved “0 kilometres over 0 hours.”

The way we make this precise is suggested by your technique in the first scenario. To avoid dividing by zero, we let time elapse just a tiny bit, then measure how much the object's position has changed in that minute interval, and perform the division. Assuming we have arbitrarily sophisticated measuring instruments, we can keep taking smaller and smaller intervals, and arrive at something closer and closer to the “instantaneous velocity.” But it is worth

stating that “instantaneous velocity” is a theoretical concept, and not one that can be measured exactly. So when you are driving in a real car, and your speedometer reads some value, say, 127 kilometres per hour, what it’s really doing is measuring your car’s change in distance over a very small period of time, and reporting to you the quotient (not unlike what you were doing in that first scenario!).

Secants and tangents. The average rate of change of a function $f(x)$ between the input x and the input $x + h$ (for some small $h > 0$) is the total amount it changes, namely $f(x + h) - f(x)$, divided by the amount the input changed, namely $(x + h) - x = h$. If you graph the function $f(x)$ on the (x, y) plane, then geometrically what you are doing is drawing a line between the points $(x, f(x))$ and $(x + h, f(x + h))$ and calculating its *slope* (the “rise over run”). Such a line cutting through the function’s graph is called a *secant line*, because it *cuts* through the function, at the very least at those two aforementioned points.

By taking h smaller and smaller (but never zero!) we continue to obtain secant lines, but the distance between the two points $(x, f(x))$ and $(x + h, f(x + h))$ will get smaller and smaller. You can imagine these points getting so close that it’s as if the *line touches the function exactly once*, at the point $(x, f(x))$ and nowhere else. This line, if it exists, will be called the *tangent line*, but we’re not ready to precisely define it just yet.

All we note at this point is that if we admit the concept of the “instantaneous rate of change,” and allow that it can be approximated by taking the the average rate of change over smaller and smaller intervals, then the slope of this tangent line (if it exists!) is exactly the instantaneous rate of change that we seek. So it seems that the slope of the tangent line should be the value of the quotient

$$\frac{f(x + h) - f(x)}{(x + h) - x} = \frac{f(x + h) - f(x)}{h}$$

when $h = 0$, if only we could plug in $h = 0$. But we cannot, since division by zero is undefined.

What we have learned from this preliminary section is that to understand rates of change, we shall have to develop a precise concept of what it means to take a function close to some input at which it is not necessarily defined. This is the notion of a *limit*.